

Discrete Differential Geometry: An Applied Introduction

Personal Notes

Mihir Rao

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1 Overview

Discrete differential geometry (DDG) studies shape from two complementary directions. Differential geometry gives a mathematical language for local quantities such as velocity, tangent direction, and bending. Geometry processing asks how to represent and compute with those quantities on finite data, especially polygonal curves and triangle meshes. The course uses each viewpoint to sharpen the other: smooth theory suggests invariants and relationships, while discrete models turn them into algorithms.

This perspective appears whenever geometry is part of the data. Examples include surface fairing in design, shape comparison, simulation, geometric learning, and analysis of biological or manufactured forms. The aim is not to memorize a catalogue of algorithms, but to recognize which mathematical description of shape is useful for a particular computation.

Definition: *Differential geometry* describes local geometric behavior and the way local behavior constrains a whole object. A typical local quantity is curvature; a typical global consequence is the total turning of a closed curve.

Definition: *Discrete differential geometry* seeks finite counterparts of those ideas. Its primitive data are quantities such as vertices, edges, lengths, angles, and areas rather than derivatives at every point.

1.1 The DDG translation problem

The same smooth object often admits several equivalent characterizations. In DDG, a productive procedure is to list those characterizations, apply each one to a finite object, and compare the resulting definitions. Equivalence in the smooth limit need not survive discretization. Consequently, there is usually no universally best discrete notion: one definition may preserve an integral identity, another may give a simple optimization problem, and a third may be more stable numerically.

This is a design principle rather than a defect. A discrete definition should be judged against the job it must do—for example, convergence under refinement, preservation of a desired invariant, or computational cost.

Note. The most natural discrete analogue of a pointwise smooth quantity is often an *integrated* quantity. Integration replaces an infinitesimal rate with a finite change over an edge, face, or other cell.

1.2 Plane curves and their local frame

Let a regular plane curve be a smooth map

$$\gamma : [0, L] \longrightarrow \mathbb{R}^2.$$

When its parameter s is arc length, $\|\gamma'(s)\| = 1$ and the unit tangent is

$$T(s) = \gamma'(s).$$

Choose the unit normal $N(s)$ by rotating $T(s)$ counterclockwise by a quarter turn. The *signed curvature* $\kappa(s)$ is defined through

$$T'(s) = \kappa(s)N(s).$$

Thus curvature is a second-order quantity: it records how the direction of a curve changes per unit length. Its sign depends on the chosen orientation. If $\theta(s)$ is the angle of the tangent, then the same information is expressed by $\kappa = d\theta/ds$.

Example: For a circle of radius r with the normal oriented toward its center, $\kappa = 1/r$. Smaller circles bend more sharply, which matches the reciprocal dependence on radius.

1.3 Why a polygon needs a new definition

Consider an oriented polygon with vertices $x_0, \dots, x_m$. Write

$$e_i = x_i - x_{i-1}, \quad \ell_i = \|e_i\|, \quad T_i = \frac{e_i}{\ell_i}.$$

Along an edge the tangent is constant, so a literal second derivative is zero. At a vertex the tangent jumps, so the same derivative is singular. A pointwise copy of smooth curvature therefore cannot describe the polygon.

The first useful replacement is the signed *turning angle* θ_i from T_i to T_{i+1} . It is a finite version of the integrated smooth identity

$$\int_a^b \kappa(s) ds = \theta(b) - \theta(a).$$

For a polygon, the curvature concentrated at vertex x_i is naturally the turning angle θ_i . This quantity has no units: it measures total turning, not bending per unit length.

Definition 1.1 [*Integrated curvature of a polygon*]. At an interior vertex x_i , the integrated curvature is the signed turning angle

$$\Theta_i = \text{atan2}(\det(T_i, T_{i+1}), \langle T_i, T_{i+1} \rangle).$$

For a consistently oriented simple closed polygon, the sum of these angles is 2π (with sign determined by orientation).

1.4 Curvature from variation of length

Curvature also controls how length changes under a normal deformation. For a smooth curve with a displacement field V , the first variation of length (ignoring endpoint terms) is

$$\delta L[V] = - \int \kappa \langle V, N \rangle ds.$$

Moving in the direction κN therefore decreases length most rapidly when the curve is free to move normally.

For a polygon, total length is $L(x) = \sum_i \ell_i$. Differentiating this finite expression with respect to an interior vertex gives

$$\nabla_{x_i} L = T_i - T_{i+1}.$$

The negative gradient $T_{i+1} - T_i$ is a *discrete curvature-normal vector*. Its magnitude is $2 \sin(|\Theta_i|/2)$, which agrees with the turning angle to first order for small angles. Unlike Θ_i , this construction arises directly from an optimization objective and is consequently convenient for gradient-based geometric flows.

1.5 Other finite viewpoints

The lecture compares two further smooth descriptions of curvature.

First, a normal offset of a smooth curve changes its length in proportion to integrated curvature. A polygon has no single normal at a vertex, so offset edges must be joined by a choice such as circular arcs, straight connectors, or intersecting extensions. These choices produce different finite length-change formulas. The ambiguity is informative: the data alone do not select one preferred discrete curvature.

Second, a smooth curve is locally approximated by an *osculating circle*, whose curvature is the reciprocal of its radius. Three consecutive polygon vertices determine a circumcircle when they are not collinear, suggesting a curvature estimate $1/R$. This estimate emphasizes local circle fitting rather than an integral law or a variational property.

Note. Turning angle, length variation, normal offsets, and circumcircles agree for a sufficiently refined smooth curve, but they are generally inequivalent on a coarse polygon. Selecting among them is part of modeling the problem.

1.6 A test case: curve shortening

Curve shortening flow evolves a curve by

$$\frac{\partial \gamma}{\partial t} = \kappa N.$$

It is a simple model of smoothing: regions of stronger bending move more quickly. In the smooth setting it offers several desirable reference properties, including preservation of total turning for a closed curve and special behavior for circles. The lecture uses it to illustrate a “no free lunch” result: a single vertex update built from a discrete curvature cannot, in general, reproduce every smooth property at once. A practical method must choose which behavior is essential for its application.

1.7 Roadmap

The remainder of the course develops this translation process for combinatorial surfaces, exterior algebra and calculus, curves, surfaces, curvature, the Laplace–Beltrami operator, conformal geometry, and geodesics. These tools provide a shared language for deriving geometry-processing algorithms rather than treating each application as an isolated recipe.

Source: Keenan Crane, Discrete Differential Geometry: An Applied Introduction, CMU 15-458/858, Lecture 1: “Overview,” official slides, <https://brickisland.net/ddg-web/lectures/DDG-Overview.pdf> (accessed August 4, 2026).